

Name	NUR ALIYA BINTI MOHD YUSRI
ID	52215125747
Group Lab	L01-B05

Metadata.

1. Extract the zip.
cd ~/Downloads
unzip image.zip

```
(kali@aliya)-[~]
$ cd ~/Downloads

(kali@aliya)-[~/Downloads]
$ unzip image.zip
Archive: image.zip
  creating: image/
  inflating: image/dog.jpg
  inflating: __MACOSX/image/._dog.jpg
  inflating: image/.DS_Store
  inflating: __MACOSX/image/._.DS_Store
  inflating: image/solitaire.exe
  inflating: __MACOSX/image/._solitaire.exe
  inflating: image/rubiks.jpg
  inflating: __MACOSX/image/._rubiks.jpg
  inflating: image/computer.jpg
  inflating: __MACOSX/image/._computer.jpg
  inflating: image/ocean.jpg
  inflating: __MACOSX/image/._ocean.jpg
```

2. ocean.jpg – Check metadata with exiftool.

exiftool ocean.jpg

```
(kali@aliya)-[~/Downloads/image]
$ exiftool ocean.jpg
ExifTool Version Number      : 11.50
File Name                    : ocean.jpg
Directory                    : .
File Size                    : 41 kb
File Modification Date/Time   : 2021:08:27 12:41:41+08:00
File Access Date/Time        : 2026:03:30 01:48:31+08:00
File Change Date/Time        : 2026:03:30 01:48:31+08:00
File Permissions              : -rwxr-xr-x
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                  : 1.01
Resolution Unit               : inches
X Resolution                  : 72
Y Resolution                  : 72
Profile CMYK Type             : Little CMS
Profile Version               : 2.1.4
Profile Class                  : Display Device Profile
Color Space Data              : RGB
Profile Connection Space      : sRGB
Profile Date Time              : 2012:01:25 03:41:57
Profile File Signature        : sRGB
Primary Platform               : Apple Computer Inc.
CM Flags                       : Not Embedded, Independent
Device Manufacturer           : 
Device Model                   : 
Device Attributes              : Reflective, Glossy, Positive, Color
Rendering Intent               : Perceptual
Connection Space Illuminant    : D50 Illuminant
Profile Creator                : 
Profile ID                     : 
Profile Description            : 
Profile Copyright              : 
Media White Point              : 0.9945 1 0.8249
Media Black Point              : 0.0298 0.0125 0.0131
Red Matrix Column              : 0.4307 0.2240 0.0192
Green Matrix Column            : 0.3555 0.7197 0.0098
Blue Matrix Column             : 0.1430 0.0061 0.7141
Red Tone Reproduction Curve    : (Binary data 64 bytes, use -b option to extract)
Green Tone Reproduction Curve  : (Binary data 64 bytes, use -b option to extract)
Blue Tone Reproduction Curve   : (Binary data 64 bytes, use -b option to extract)
Comment                       : THIS IS THE HIDDEN FLAG
Image Width                   : 4096
```

exiftool ocean.jpg | grep -i "comment\|flag\|CTF"

```
(kali@aliya)-[~/Downloads/image]
$ exiftool ocean.jpg | grep -i "comment\|flag\|CTF"
CMM Flags      : Not Embedded, Independent
Comment        : THIS IS THE HIDDEN FLAG
```

3. computer.jpg – Look for hidden strings.
`strings computer.jpg`
`strings computer.jpg | grep -i "flag\[CTF\]hidden"`
`hexeditor computer.jpg`
4. dog.jpg – Extract embedded files with binwalk.
`binwalk dog.jpg`
`binwalk -e dog.jpg`
`cd _dog.jpg.extracted/`
`ls`
`cat hidden_text.txt`

```
(kali@kaliya)-[~/Downloads/image]
$ binwalk dog.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, JFIF standard 1.01
88221	0x1589D	Zip archive data, at least v1.0 to extract, compressed size: 21, uncompressed size: 21, name: hidden_text.txt
88384	0x15940	End of Zip archive, footer length: 22

```
(kali@kaliya)-[~/Downloads/image]
$ binwalk -e dog.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
88221	0x1589D	Zip archive data, at least v1.0 to extract, compressed size: 21, uncompressed size: 21, name: hidden_text.txt

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented

```
(kali@kaliya)-[~/Downloads/image]
$ cd _dog.jpg.extracted/

(kali@kaliya)-[~/Downloads/image/_dog.jpg.extracted]
$ ls
1589D.zip  hidden_text.txt

(kali@kaliya)-[~/Downloads/image/_dog.jpg.extracted]
$ cat hidden_text.txt
THIS IS A HIDDEN FLAG
```

5. rubiks.jpg – Check file type first, then binwalk.
`file rubiks.jpg`
`binwalk rubiks.jpg`
`binwalk -e rubiks.jpg`

```
(kali@kaliya)-[~/Downloads/image]
$ file rubiks.jpg
rubiks.jpg: PNG image data, 609 x 640, 8-bit/color RGBA, non-interlaced

(kali@kaliya)-[~/Downloads/image]
$ binwalk rubiks.jpg
```

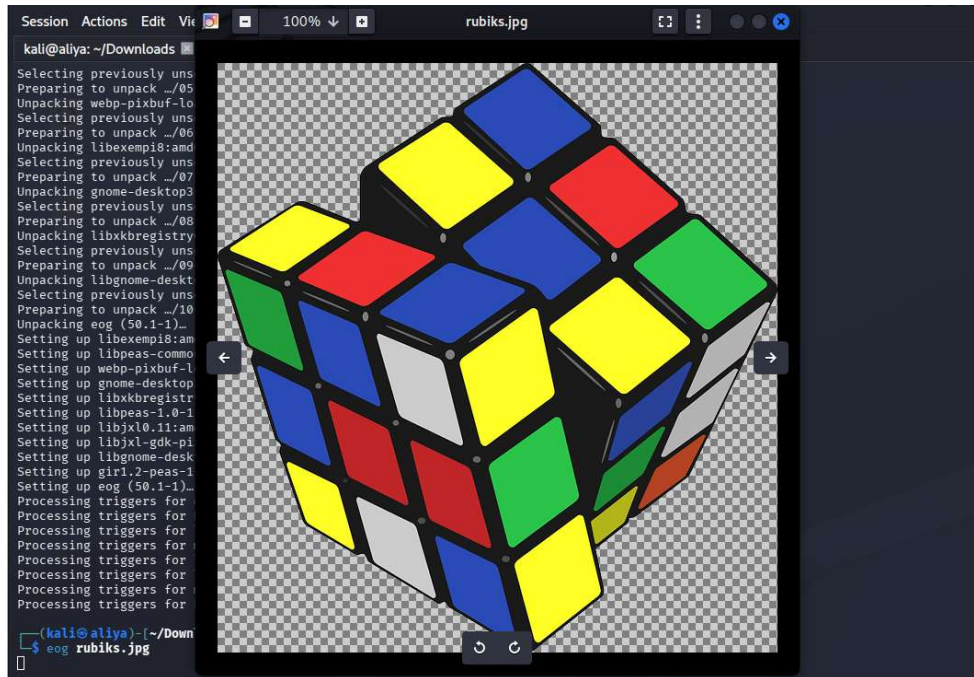
DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	PNG image, 609 x 640, 8-bit/color RGBA, non-interlaced
41	0x29	Zlib compressed data, best compression

```
(kali@kaliya)-[~/Downloads/image]
$ binwalk -e rubiks.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
41	0x29	Zlib compressed data, best compression

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented

eog rubiks.jpg



6. solitaire.exe – An image disguised as .exe

file solitaire.exe

strings solitaire.exe

binwalk -e solitaire.exe

```
(kali@aliya) - [~/Downloads/image]
$ file solitaire.exe
solitaire.exe: PNG image data, 640 x 449, 8-bit/color RGBA, non-interlaced

(kali@aliya) - [~/Downloads/image]
$ strings solitaire.exe
IHDR
IDATx
w|TU
A]]uW
NII
8Nw!D&
4MsU
<ZZZ
iH)Q
GAAa7
PAawB<
;v4^~
qRRR
'UWW
bR2u

(kali@aliya) - [~/Downloads/image]
$ binwalk -e solitaire.exe

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
41           0x29           Zlib compressed data, best compression

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented
```